

COMSATS University Islamabad, Vehari Campus Pakistan

Department of Computer Science

Course: CSC336 – Web Technologies Lab Assignment 2

GitHub Repository Link:

<https://github.com/ahsanali/Web-Tech-Lab-Ass-02>



**Building a Laravel-Based User Management System
with Full CRUD Functionality, Profile Image Handling,
and Email Search Feature**

by

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Abstract

This report presents the design and implementation of a web-based User Management System developed using the Laravel PHP framework. The system is built following the Model–View–Controller (MVC) architectural pattern and provides a complete suite of CRUD (Create, Read, Update, Delete) operations for managing user records. Key features include profile image upload and storage, email-based filtering of records, and input validation with confirmation dialogs to prevent accidental data loss. The application is deployed locally using Laravel’s built-in Artisan development server and uses MySQL as the backend database. The project demonstrates practical application of Laravel routing, Eloquent ORM, Blade templating, and form handling in a real-world web development scenario.

Keywords: Laravel, CRUD, MVC, PHP, User Management, Image Upload, Search, Blade Templates, Eloquent ORM, MySQL.

List of Abbreviations

Abbreviation	Full Form
CRUD	Create, Read, Update, Delete
MVC	Model–View–Controller
PHP	Hypertext Preprocessor
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
SQL	Structured Query Language
URL	Uniform Resource Locator
ORM	Object Relational Mapping
IDE	Integrated Development Environment
UI	User Interface
DB	Database
HTTP	Hypertext Transfer Protocol
REST	Representational State Transfer
CMD	Command Prompt / Terminal

Chapter 1

Introduction and System Overview

1.1 Background and Motivation

Managing user data is a fundamental requirement in virtually every modern web application — ranging from academic portals and HR systems to e-commerce platforms and administrative dashboards. Traditional methods of record keeping, such as spreadsheets or paper-based forms, are inefficient, error-prone, and difficult to scale. Web-based systems built on robust frameworks provide a structured, maintainable, and scalable alternative.

This project, **Building a Laravel-Based User Management System with Full CRUD Functionality, Profile Image Handling, and Email Search Feature**, is designed to address these challenges by implementing a complete user record management solution using the Laravel framework. Laravel’s expressive syntax, built-in ORM (Eloquent), and Blade templating engine make it an ideal choice for rapid development of data-driven web applications.

The system enables authorised users to register new records, browse all saved entries in a structured table, update existing information, and permanently remove records after explicit confirmation. Profile images can be attached to each user record, and a real-time email search allows quick retrieval of specific entries from potentially large datasets.

1.2 Tools and Technologies

Table [1.1](#) lists the tools and technologies selected for the development of this project. Each component was chosen based on compatibility, community support, and suitability for Laravel-based web development.

1.3 Project Objectives

The core objectives guiding the design and development of this system are:

- **PO-1:** Design and implement a fully functional web application that supports Create, Read, Update, and Delete operations on user records using the Laravel MVC framework.

Table 1.1: Tools and Technologies Used

Tool / Technology	Version	Purpose
Laravel	Latest (v11.x)	Primary PHP framework; implements MVC architecture and handles routing, ORM, and views.
PHP	8.x	Server-side scripting language required to execute Laravel applications.
Composer	Latest	PHP dependency manager used to install and manage Laravel packages.
MySQL	Latest	Relational database used to persist user records, including image file paths.
Blade	Built-in	Laravel's templating engine used to build dynamic HTML views and forms.
Bootstrap 5	5.x	Frontend CSS framework used to create a responsive, clean user interface.
VS Code	Latest	Primary code editor used for writing and debugging the application.
Git & GitHub	Latest	Version control and remote repository hosting for the project codebase.

- **PO-2:** Extend the base CRUD system with additional usability features — specifically, profile image upload and display, email-based record search, and input validation with user-confirmation prompts for destructive actions.
- **PO-3:** Follow clean code and architectural conventions (MVC pattern, RESTful routing, Eloquent ORM) to produce a maintainable and extensible codebase.

Chapter 2

Implementation

2.1 Development Environment

The project was built and tested on a local Windows machine using the tools listed in Table 2.1.

Table 2.1: Development Environment Setup

Component	Tool / Technology	Role
Backend Framework	Laravel (v11.x)	MVC architecture, routing, ORM, validation.
Programming Lang.	PHP 8.x	Server-side execution of Laravel code.
Dependency Mgr.	Composer	Installs Laravel and all required packages.
Database	MySQL / SQLite	Stores user records and image file paths.
View Engine	Blade + Bootstrap 5	Dynamic UI pages: forms, tables, alerts.
Code Editor	Visual Studio Code	Writing, debugging, and managing files.
Local Server	Laravel Artisan	Runs the application via <code>php artisan serve</code> .
Version Control	Git + GitHub	Tracks code changes and hosts the repository.

2.2 Project Setup and Execution

To run the project on a local machine, execute the following steps in the terminal from inside the project root directory:

Listing 2.1: Project Setup Commands

```
1 # Step 1      Install all PHP dependencies
2 composer install
3
4 # Step 2      Copy environment file and set the database
5               credentials
6 cp .env.example .env
```

```
6 # Edit .env: DB_DATABASE=your_db DB_USERNAME=root DB_PASSWORD=
7
8 # Step 3      Generate application key
9 php artisan key:generate
10
11 # Step 4      Run database migrations to create the table
12 php artisan migrate
13
14 # Step 5      Create the uploads directory (if not present)
15 mkdir public/uploads
16
17 # Step 6      Start the local development server
18 php artisan serve
```

After the server starts, open `http://127.0.0.1:8000` in any browser to access the application.

2.3 Core Implementation Details

2.3.1 CRUD Operations

All CRUD operations are implemented through Laravel's resourceful routing and a dedicated `UserController`. The key controller methods are:

- **index()**: Retrieves all user records (with optional email filter) and passes them to the `users/index` Blade view.
- **create()**: Returns the `users/create` registration form view.
- **store()**: Validates incoming form data, handles the image upload, and persists a new record to the database.
- **edit(\$id)**: Retrieves the specified user record and populates the edit form view.
- **update(\$id)**: Validates and saves changes to the selected user record; replaces the image if a new file is uploaded.

UserManager

MAIN MENU

Register user

All users

Settings

Register new user

Fill all required fields to create a user record

PERSONAL INFORMATION

Full name *

Ahsan Raza

Email address *

ahsan@example.com

CNIC number *

35202-1234567-1

Telephone *

+92-300-1234567

Comments

Senior developer — Rahim Yar Khan branch.

PROFILE PHOTO

Click to upload or drag & drop · PNG, JPG up to 2MB

Cancel

Save user

Register user

All users

Settings

User registered successfully!

TOTAL USERS

24

THIS MONTH

6

WITH PHOTO

18

WITHOUT PHOTO

6

Search

Clear

#	USER	CNIC	TELEPHONE	PHOTO
1	Ahsan Raza ahsan@example.com	35202-1234567-1	+92-300-1234567	Yes
2	Sara Baig sara.baig@mail.com	35202-9876543-2	+92-321-9876543	Yes
3	Mohsin Khan mohsin.k@gmail.com	35202-5555555-3	+92-333-5551234	No
4	Fatima Shah fatima.s@example.com	35202-1122334-4	+92-311-1122334	Yes

<

1

2

3

>

UserManager

MAIN MENU

Register user

All users

Settings

Edit user

Update details and save changes

Editing ID #1

AD

PERSONAL INFORMATION

Full name *

Ahsan Raza

Email address *

ahsan@example.com

CNIC number *

35202-1234567-1

Telephone *

+92-300-1234567

Comments

Senior developer — Rahim Yar Khan branch.

PROFILE PHOTO

AR

Current photo
profile_ahsan.jpg — upload a new file to replace

Upload new photo (optional)

← Back

Update user

UserManager

MAIN MENU

Register user

All users

Settings

Register new user

Fix the errors below before submitting

3 validation errors found

The email field is required.

CNIC must match format XXXXX-XXXXXXXX-X.

Telephone must be a valid number.

PERSONAL INFORMATION

Full name *

Ahsan Raza

Email address * ⓘ

The email field is required.

CNIC number * ⓘ

35202-ABC

Format: XXXXX-XXXXXXXX-X

Telephone * ⓘ

abc-xyz

Must be a valid phone number.

Comments

Senior developer...

Cancel

Save user

UserManager

MAIN MENU

Register user

All users

Settings

All users

Manage registered users

Delete this user?

You are about to permanently delete **Ahsan Raza** and all associated data. This cannot be undone.

Conclusion

This lab assignment successfully demonstrates the practical implementation of a web-based User Management System using the Laravel PHP framework. The application covers all four CRUD operations in a clean, maintainable codebase structured according to the MVC design pattern. Additional features — email-based search, profile image upload and display, server-side input validation, and JavaScript confirmation dialogs — extend the system beyond a basic data-entry tool into a usable, real-world application component.

Key learning outcomes achieved through this project include:

- Hands-on experience with Laravel routing, controllers, models, and Blade views.
- Practical understanding of Eloquent ORM for database interactions.
- Implementation of file upload handling in a PHP/Laravel context.
- Application of server-side validation and flash messaging for user feedback.
- Use of Git and GitHub for version control throughout the development cycle.

The project can be extended in future work by adding user authentication, role-based access control, pagination, export to CSV/PDF, and deployment to a cloud hosting platform.